

Tobacco smoking and chewing, alcohol drinking and lung cancer risk among men in southern India.

In India, lung cancer is one of the most common and lethal cancers, and tobacco smoking remains its most important etiologic factors. The objective of our study is to examine the effects of different tobacco consumption forms, including smoking and chewing, on lung cancer risk of men in southern India, especially to compare the effects of bidi smoking to cigarette smoking on lung carcinogenesis. We also evaluated the possible role of Indian alcohol beverages and non-Indian alcohol beverages on lung carcinogenesis. We conducted a case-control study in Chennai and Trivandrum. In total, 778 lung cancer cases and 3,430 controls, including 1,503 cancer controls and 1,927 healthy controls, were recruited. The effects of cigarette, bidi smoking, chewing and alcohol drinking on the risk of lung cancer were estimated from unconditional multivariate logistic regression. We also applied the generalized additive model (GAM) with locally-weighted running-line smoothers (loess) to find the most plausible curve for the dose-response relationship. The results from GAM suggest a plateau after 35 years of smoking or 10 cigarette-equivalent pack-years for both cigarette and bidi. The OR is 4.54 (95%CI=2.96-6.95) and 6.45 (95%CI=4.38-9.50) for more than 30 years of cigarette-only and bidi-only smoking, respectively, and 6.87 (95%CI=4.62-10.2) and 10.7 (95%CI=5.82-19.6) for more than 12 weighted cumulative cigarette-only and bidi-only consumption, respectively. The lung cancer risk of former cigarette smokers drops down more quickly after quitting smoking compared to former bidi smokers. There is no evidence for the effect of chewing and lung cancer risk nor clear evidence of an effect of overall alcohol drinking among never-smokers, although Indian alcohol drinking seemed to remain associated with lung cancer risk under limited power (OR=2.67, 95%CI=1.02-7.02). Bidi smoking seems to have a stronger carcinogenic effect than cigarette smoking; this difference holds no matter which aspect of smoking was considered.

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